

ROB PATRICK

LEAD AI/ML ENGINEER

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Objective

Lead AI/ML Engineer with 10+ years of experience architecting, developing, and deploying large-scale Generative AI and Machine Learning solutions. Skilled in designing LLM-based systems, multi-agent frameworks, RAG pipelines, model fine-tuning, and conversational AI platforms. Proficient in LangChain, LangGraph, Hugging Face, Vector Databases, LoRA/QLoRA, FastAPI, and GPU-optimized model serving. Possesses a strong foundation in NLP, Computer Vision, deep learning, automation, and voice-AI technologies, with a proven track record of leading end-to-end AI initiatives from concept to production.

Experience

Lead AI/ML Engineer | New Era Technology Oct 2022 - Present

- Built enterprise-grade GenAI systems integrating LLM agents + RAG + Vector Search.
- Architected multi-agent orchestration using LangChain + LangGraph + MCP tools.
- Designed and deployed **voicebots & enterprise chatbots** for automated client interaction.
- Automated workflows using **n8n / Zapier / Make.com**, connecting AI models to business pipelines.
- Achieved **22% improvement** in AI task accuracy via LangSmith evaluation pipelines.
- Implemented RAG pipelines with Pinecone & QLoRA-fine-tuned models for cost-efficient inference.
- Built AI-powered trading strategy analysis & backtesting engine using Flask + LLMs.
- Built plant disease Vision AI (Transfer learning - VGG/ResNet) on AWS SageMaker.
- Fine-tuned & containerized Wave2Lip for scalable GPU workloads on RunPod.

Senior Machine Learning Engineer | Integrity Power Mar 2018 – Sept 2022

- Developed Azure ML predictive analytics system for time-series forecasting.
- Built automated ASR-based voice data pipelines and conversational NLP systems.
- Created custom NER pipeline for structured document extraction.
- Trained masked language models for intelligent text automation.
- Developed ML infrastructure for deployment, monitoring & scaling.

Associate Data Scientist | Radical Imaging Mar 2015 – Feb 2018

- Delivered CV + NLP models for clinical automation.
- Optimized CNN + LSTM fall-detection (1.5s → 0.1s latency).
- Built YOLO-based hazard detection (95% accuracy).
- Created synthetic data generator for medical AI.
- Migrated legacy desktop tools into modern web-based modular AI platform.

Education

MS Computer Science | PUCIT

Skills

- **PROGRAMMING LANGUAGES:**
Python, C++, R, SQL
- **FRAMEWORKS & LIBRARIES:**
LangChain, LangGraph, HuggingFace, Transformers, OpenAI API, MCP (Model Context Protocol), Vector DBs (Pinecone, Weaviate, ChromaDB), PyTorch, TensorFlow, NumPy, Pandas, Scikit-Learn, FastAPI, Flask, SpaCy, NLTK, Librosa, OpenCV, YOLO, Keras, CUDA
- **GENAI & MLOps:**
QLoRA, PEFT, Model Fine-tuning, RAG Pipelines, Multi-Agent AI Systems, Conversational AI, Voicebot Development, Chatbot Orchestration, LangSmith, Prompt Engineering, Docker, Kubernetes, GitLab/Jenkins CI/CD, MLflow, DVC, Runpod, ONNX Runtime.
- **DATABASES:**
PostgreSQL, MySQL, MS SQL Server, MongoDB, Redis
- **CLOUD PLATFORMS:**
AWS (SageMaker, Lambda, EC2, S3, RDS), GCP AI Services, Azure ML, Docker
- **TOOLS:**
Jupyter, Jira, Trello, Spark, Linux, Git, n8n, Zapier, [Make.com](#)

Projects

- Enterprise RAG System — LangGraph, Multi-LLM routing, Pinecone, QLoRA.
- AI Research Agent — MCP tool calling, document QA, auto-knowledge extraction.
- Financial LLM Engine — Automated financial insights & recommendations.
- Medical Claims NLP Engine — OCR + NER + Claim decision models.
- Voicebot and Chatbot Platform — LLM + ASR + TTS + n8n automation.
- Edge AI Vision Pipeline — YOLO, transfer learning, real-time inference.